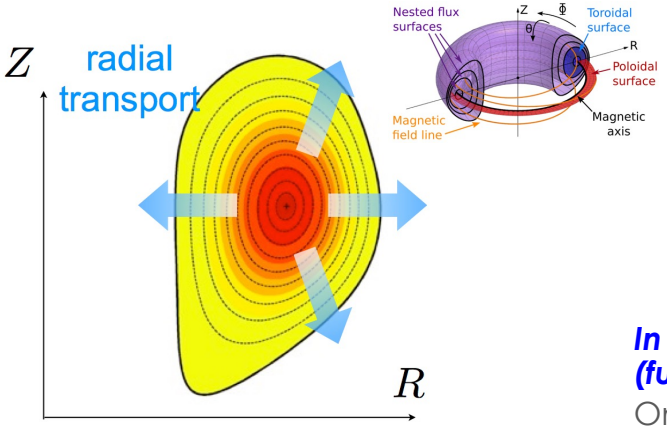
Plasma transport and the meaning of "fluxes"



Flux = amount crossing a flux surface per area per time (outward > 0; from core to the wall)

$$\frac{\partial\langle n_a\rangle}{\partial t}+\frac{1}{V'}\frac{\partial}{\partial r}(V'\Gamma_a)=S_{na}$$

$$\frac{\partial\langle n_a T_a\rangle}{\partial t}+\frac{1}{V'}\frac{\partial}{\partial r}(V'Q_a)=S_{Ta}$$

Particle flux Γ_a sets the density profile n_a

Energy (Heat) flux Q_a sets the temperature profile T_a

In steady-state: Net flow (flux) out through the surface balances the sources S (fueling, heating) inside it.

One pair of equations per species a (electrons, hydrogen ions, impurities).

- The fluxes **Γ** and **Q** come from turbulence. They are computed by codes (**CGYRO or TGLF**). They depend on n, T, local gradients (dn/dr, dT/dr), many other parameters

- **In our data: out.tglf.gbflux**
  - **Gamma_1, Gamma_2** → Γ_a
  - **Q_1, Q_2** → Q_a
  - **Pi_1, Pi_2** → Π_a (momentum flux)